

# HTML

## 1. History Of HTML:

- HTML stands for **H**yper **T**ext **M**arkup **L**anguage.
- Hypertext refers to the way in which Web pages (HTML documents) are linked together.
- HTML is markup language which means we use HTML to simply “mark-up” a text document with tags that tell a web browser how to structure it to display.
- HTML is standard markup language for creating web pages.
- It is developed by Tim Berners Lee in 1990.



- HTML describes the structure of web pages using markup.
- HTML elements are building blocks of HTML pages.
- HTML elements are represented by tags.

## 2. Versions of HTML:

| Versions  | Year |
|-----------|------|
| HTML      | 1991 |
| HTML 2.0  | 1995 |
| HTML 3.2  | 1997 |
| HTML 4.01 | 1999 |
| XHTML     | 2000 |
| HTML5     | 2014 |

### 3. HTML Tags:

- HTML tags normally come in pairs like `< >` and `</>`.
- The first tag in a pairs is the start tag ,the second tag is the end tag.
- End tag is written like the start tag but with a forward slash(/) inserted before the tag name.
- **Syntax :** `<tagname>content goes here....</tagname>`.
- HTML tags are ordered by different categories such as
  - 3.1 Basic HTML.
  - 3.2 Formatting.
  - 3.3 Forms and Input.
  - 3.4 Frames.
  - 3.5 Images.
  - 3.6 Audio/Video.
  - 3.7 Links.
  - 3.8 Lists.
  - 3.9 Tables.
  - 3.10 Styles and Semantics.
  - 3.11 Programming.

#### 3.1 Basic HTML Tags :

- **<!DOCTYPE>** : Defines the type of document type and the version of html we are using. It must appear only once , at the top of the page.
- **<html>** : Defines an HTML document.
- **<head>** : Defines information about the document.
- **<title>** : Defines a title for the document.
- **<body>** : Defines the documents body.
- **<!-- .... -->** : Defines a comment.

#### A Simple HTML Document:

```
<!DOCTYPE html>
<html lang="en">
  <head>  <meta charset="UTF-8">
    <title>Document</title>
  </head>
  <body>
    <h1>Simple HTML</h1>
  </body>  </html>
```

**Meta Tag:**

- Metadata is data (information) about data.
- The <meta> tag provides metadata about the HTML document. Metadata will not be displayed on the page, but will be machine parsable.
- Meta elements are typically used to specify page description, keywords, author of the document, last modified, and other metadata.
- The metadata can be used by browsers (how to display content or reload page), search engines (keywords), or other web services.
- HTML5 introduced a method to let web designers take control over the viewport (the user's visible area of a web page), through the <meta> tag.
- <meta> tags always go inside the <head> element.
- Metadata is always passed as name/value pairs.
- The content attribute MUST be defined if the name or the http-equiv attribute is defined. If none of these are defined, the content attribute CANNOT be defined.
- The viewport is the user's visible area of a web page. It varies with the device, and will be smaller on a mobile phone than on a computer screen.
- You should include the following <meta> viewport element in all your web pages:  

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```
- A <meta> viewport element gives the browser instructions on how to control the page's dimensions and scaling.
- The width=device-width part sets the width of the page to follow the screen-width of the device (which will vary depending on the device).
- The initial-scale=1.0 part sets the initial zoom level when the page is first loaded by the browser.

**Without the viewport meta tag****With the viewport meta tag**

### • Examples

- ❖ Example 1 - Define keywords for search engines:

```
<meta name="keywords" content="HTML, CSS, XML, XHTML, JavaScript">
```

- ❖ Example 2 - Define a description of your web page:

```
<meta name="description" content="Free Web tutorials on HTML and CSS">
```

- ❖ Example 3 - Define the author of a page:

```
<meta name="author" content="John Doe">
```

- ❖ Example 4 - Refresh document every 30 seconds:

```
<meta http-equiv="refresh" content="30">
```

- ❖ Example 5 - Setting the viewport to make your website look good on all devices:

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

### • Attributes

| Attribute         | Value                                                                          | Description                                                                                              |
|-------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <u>charset</u>    | <i>character_set</i>                                                           | Specifies the character encoding for the HTML document                                                   |
| <u>content</u>    | <i>text</i>                                                                    | Gives the value associated with the http-equiv or name attribute                                         |
| <u>http-equiv</u> | content-type<br>default-style<br>refresh                                       | Provides an HTTP header for the information/value of the content attribute                               |
| <u>name</u>       | application-name<br>author<br>description<br>generator<br>keywords<br>viewport | Specifies a name for the metadata                                                                        |
| <u>scheme</u>     | <i>format/URI</i>                                                              | Not supported in HTML5.<br>Specifies a scheme to be used to interpret the value of the content attribute |

- Example for Meta tag charset.

```
<meta charset="UTF-8">
```

**UTF-8** is a variable width character encoding capable of encoding all 1,112,064 valid code points in Unicode using one to four 8-bit bytes. The encoding is defined by the Unicode standard, and was originally designed by Ken Thompson and Rob Pike. The name is derived from Unicode (or Universal Coded Character Set) Transformation Format – 8-bit.

**Marquee Tag:**

- An HTML marquee is scrolling piece of text displayed either horizontally across, or vertically down on the webpage based on the settings. This is created by using HTML `<marquee>` tag.
- In HTML 5 this tag is deprecated (not used).

- **Syntax:**

```
<marquee attribute _name="attribute_value".....more attributes>
    Text message or image
</marquee>
```

- The `<marquee>` tag uses different attributes and are as follows:
  - **Width:** Specifies width of the marquee. This can be value 10 or 20% etc.
  - **Height:** Specifies height of the marquee. This can be value 10 or 20%etc.
  - **Direction:** Specifies direction in which marquee should scroll. This can be a value like up, down, left or right.
  - **Behavior:** Specifies the type of scrolling of the marquee. This can have value like scroll, slide and alternate.
  - **Scrolldelay:** Specifies how long to delay between each jump. This will have a value like 10 sec.
  - **Scrollamount:** Specifies speed of the marquee text. This can have a value like 10 etc.
  - **Loop:** Specifies how many times to loop. The default value is INFINITE which means that marquee loops endlessly.
  - **Bgcolor:** Specifies background color in terms of color name or color hex value.
  - **Hspace:** Specifies horizontal space around the marquee. This can be value like a 10 or 20%etc.
  - **Vspace:** Specifies vertical space around the marquee. Thi can be value like a 10 or 20%etc.

- **Example:** `<!DOCTYPE html>`

```
<html>  <head>
    <title>HTML Marquee Tag</title>
</head>
<body>
    <marquee>This is example of marquee</marquee>
</body>
</html>
```

**Heading Tags:**

- Any documents in html start with a heading. We can use different sizes of headings.
- HTML uses six levels of headings, which use the elements <h1>, <h2>, <h3>, <h4>, <h5>, and <h6>.
- While displaying any heading, browser adds one line before and one line after that heading.
- **Example:**

```
<!DOCTYPE html>
<html>
  <head>
    <title>Heading Example</title>
  </head>
  <body>
    <h1>This is heading 1</h1>
    <h2>This is heading 2</h2>
    <h3>This is heading 3</h3>
    <h4>This is heading 4</h4>
    <h5>This is heading 5</h5>
    <h6>This is heading 6</h6>
  </body>
</html>
```

**Paragraph Tag:**

- The <p> tag defines a way to structure text into different paragraphs.
- Each paragraph of text should go in between an opening <p> and a closing </p> tag.
- **Example:**

```
<!DOCTYPE html>
<html>
  <head>
    <title>Paragraph Example</title>
  </head>
  <body>
    <p>Here is a first paragraph of text.</p>
    <p>Here is a second paragraph of text.</p>
  </body>
</html>
```

**Line Break Tag:**

- The `<br />` tag is used to make space between the characters **br** and the forward slash.
- Whenever we use `<br/>` element, anything following it starts from the next line.
- This tag is an example of an **empty** element, where we do not need opening and closing tags, as there is nothing to go in between them.

- **Example:**

```
<!DOCTYPE html>
<html>
  <head>
    <title>Line Break Example</title>
  </head>
  <body>
    <p> Hello<br /> Good Morning<br /> Have a nice day.<br
  /></p>
  </body>
</html>
```

**Horizontal Lines:**

- Horizontal tags are used to break sections of a document.
- The `<hr />` tag creates a line from the current position in the document to the right margin and breaks the line accordingly.
- The `<hr />` tag is an example of the empty element, where there is no need of opening and closing tags.

- **Example:**

```
<!DOCTYPE html>
<html>
  <head>
    <title>Horizontal Line Example</title>
  </head>
  <body>
    <p>This is paragraph one.</p>
    <hr />
    <p>This is paragraph second.</p>
  </body>
</html>
```

**Preserve Formatting:**

- Sometimes, we want our text to follow the exact format of how it is written in the HTML document. In such cases, will use preformatted tag <pre>.
- Any text between the opening <pre> tag and the closing </pre> tag will preserve the formatting of the source document.

- **Example:**

```
<!DOCTYPE html>

<html> <head>
    <title>Preserve Formatting Example</title>
</head>
<body>
    <pre>
        Function testFunction (strText) {
            Alert (strText)
        }
    </pre>
</body>
</html>
```

**Nonbreaking Spaces:**

- In cases, where we do not want the client browser to break text, we should use a nonbreaking space entity &nbsp; instead of a normal space.

- **Example:**

```
<!DOCTYPE html>
<html>
    <head>
        <title>Nonbreaking Spaces Example</title>
    </head>
    <body>
```

example of this technique appears in the movie  
“12  Angry Men.”

```
</p>
</body>
</html>
```

**Strong Tag:**

- The HTML **<strong>** tag gives text a strong emphasis which traditionally means that the text is displayed as bold by the browser. This tag is also commonly referred to as the **<strong>** element.
- **Example :**

```
<!DOCTYPE html>
<html>
  <head>
    <title>Strong Tag Example</title>
  </head>
  <body>
    <p>Good Morning <strong>Have a Nice Day </strong></p>
  </body>
</html>
```

**Bold Tag:**

- The HTML **<b>** element defines **bold** text, without any extra importance.

- **Example:** **<!DOCTYPE html>**

```
<html>
  <head>
    <title>Bold Tag Example</title>
  </head>
  <body>
    <p>Good Morning <b>Have a Nice Day </b></p>
  </body>
</html>
```

**Span Tag:**

- The HTML <span> tag is used for grouping and applying styles to inline elements.
- There is a difference between the span tag and the div tag. The span tag is used with inline elements whilst the div tag is used with block-level content.

- **Example:**

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML span Tag</title>
  </head>
  <body>
    <p>This is a paragraph <span style = "color:#FF0000;">
      This is a paragraph</span>This is a paragraph</p>
  </body>
</html>
```

**Sup and Sub Tag:**

- The <sup> tag defines superscript text. Superscript text appears half a character above the normal line, and is sometimes rendered in a smaller font.
- The <sub> tag defines subscript text. Subscript text appears half a character below the normal line, and is sometimes rendered in a smaller font.

- **Example:**

```
<!DOCTYPE html>
<html>
  <head>
    <title>Sup and Sub Tag</title>
  </head>
  <body>
    <p>This text contains <sub>subscript</sub> text.</p>
    <p>This text contains <sup>superscript</sup> text.</p>
  </body>
</html>
```

The attributes listed below are supported by almost all the html 5 tags.

| Attribute       | Options                          | Function                                                                                                |
|-----------------|----------------------------------|---------------------------------------------------------------------------------------------------------|
| accesskey       | User Defined                     | Specifies a keyboard shortcut to access an element.                                                     |
| Align           | right, left, center              | Horizontally aligns tags                                                                                |
| background      | URL                              | Places an background image behind an element                                                            |
| Bgcolor         | numeric, hexadecimal, RGB values | Places a background color behind an element                                                             |
| Class           | User Defined                     | Classifies an element for use with Cascading Style Sheets.                                              |
| contenteditable | true, false                      | Specifies if the user can edit the element's content or not.                                            |
| contextmenu     | Menu id                          | Specifies the context menu for an element.                                                              |
| data-XXXX       | User Defined                     | Custom attributes. Authors of a HTML document can define their own attributes. Must start with "data-". |
| draggable       | true,false, auto                 | Specifies whether or not a user is allowed to drag an element.                                          |
| Height          | Numeric Value                    | Specifies the height of tables, images, or table cells.                                                 |
| Hidden          | hidden                           | Specifies whether element should be visible or not.                                                     |
| Id              | User Defined                     | Names an element for use with Cascading Style Sheets.                                                   |
| Item            | List of elements                 | Used to group elements.                                                                                 |
| Itemprop        | List of items                    | Used to group items.                                                                                    |

|            |                     |                                                                     |
|------------|---------------------|---------------------------------------------------------------------|
| Spellcheck | true, false         | Specifies if the element must have its spelling or grammar checked. |
| Style      | CSS Style sheet     | Specifies an inline style for an element.                           |
| Subject    | User define id      | Specifies the element's corresponding item.                         |
| TabIndex   | Tab number          | Specifies the tab order of an element.                              |
| Title      | User Defined        | "Pop-up" title for your elements.                                   |
| Valign     | top, middle, bottom | Vertically aligns tags within an HTML element.                      |
| Width      | Numeric Value       | Specifies the width of tables, images, or table cells.              |

## 3.2 Formatting Tags:

### Centering Content:

- We can use **<center>** tag to put any content in the center of the page or any table cell.
- **Example:**

```
<!DOCTYPE html>
<html>
    <head>
        <title>Centering Content Example</title>
    </head>
    <body>
        <center>
            <p>This text is in the center.</p>
        </center>
    </body>
</html>
```

### Font Tag:

- Defines font, color, and size for text **<font>** tag is not supported in HTML 5.
- **Example:**

```
<!DOCTYPE html>
<html>
    <body>
        <p><font size="3" color="red">This is some text</font></p>
        <p><font face="verdana" color="green">This is some text</font></p>
        <p>The font element is not supported in HTML5.Use CSS instead</p>
    </body>
</html>
```

### 3.3 Forms and Input Tags:

#### Form Tag:

- Defines an HTML form that is used to collect user input. An HTML **<form>** contains form elements. Form elements are different types of input elements, like text fields, checkboxes, radio buttons, submit buttons and more.

**Syntax:** <form>

Form elements

</form>

#### Input Tag:

- Defines an input control. **<input>** element can be displayed in several ways, depending on the **type** attribute.

<input type="text" > ---- Defines an one-line text field.

<input type="radio"> ---- Defines a radio button(selecting one of many choices.

<input type="submit"> --- Defines a submit button(submitting the form).

- **Example: Text Input**

```
<!DOCTYPE html>
```

```
<html> <body>
```

```
<form>
```

```
  FirstName: <input type="text" name="firstname">
```

```
  LastName:<input type="text" name="lastname">
```

```
</form>
```

```
</body>
```

```
</html>
```

- **Example: Radio Button Input**

```
<!DOCTYPE html>
```

```
<html> <body>
```

```
<form>
```

```
<input type="radio" name="gender" value="male" checked>Male<br>
```

```
<input type="radio" name="gender" value="female"><br>
```

```
<input type="radio" name="gender" value="other">
```

```
</form>
```

```
</body>
```

```
</html>
```

- **Example: The Submit Button**

- `<input type="submit">` defines a button for submitting the form data to a form handler. A form-handler is a server page with script for processing Input data. The form-handler is specified in the form's action attribute.

```
<html> <body>
    <form action="action_page.php">
        FirstName: <input type="text" name="firstname">
        LastName:<input type="text" name="lastname">
        <input type="submit" name="submit" value="submit">
    </form>
</body>
</html>
```

- **The Action Attribute:**

- The **action** attribute defines the action to be performed when the form is submitted.
- Normally the form data is sent to a web page on the server when the user clicks on the submit button.
- In the example above, the form data is sent to a page on the server called "action\_page.php".
- This page contains a server-side script that handles the form data.
- `<form action="/action_page.php">` if the action attribute is omitted, the action is set to the current page.

- **The Target Attribute:**

- The **target** attribute specifies if the submitted result will open in a new browser tab, a frame, or in the current window.
- The default value is "`_self`" which means the form will be submitted in the current window.
- To make the form result open in a new browser tab, use the value "`_blank`":

- **Example:**

```
<html> <body> <form action="action_page.php" target="_blank">
    Firstname:<input type="text" name="firstname">
    Lastname:<input type="text" name="lastname">
    <input type="submit" name="submit" value="submit">
</form>
</body>
</html>
```

- **The Method Attribute:**

- The method attribute specifies the HTTP method (GET or POST) to be used when submitting the form data:

- **Example:**

```
<html>
  <body>
    <form action="action_page.php" target="_blank" method="get">
      Firstname:<input type="text" name="firstname">
      Lastname:<input type="text" name="lastname">
      <input type="submit" name="submit" value="Submit">
    </form>
  </body>
</html>
```

### When to use GET?

- The default method when submitting the form data is GET.
- However, when GET is used, the submitted form data will be visible in the page address field like /action\_page.php?firstname=jhon &lastname=harry.
- Appends form-data into the URL in name/value pairs.
- The length of a URL is limited(about 3000 characters).
- GET cannot be used to send sensitive data(visible in the URL).
- Useful for form submissions where a user want to bookmark the result.
- GET is better for non-secure data, like query strings in Google.

**Example:**

```
<html>
  <body>
    <form action="action_page.php" target="_blank" method="post">
      Firstname:<input type="text" name="firstname">
      Lastname:<input type="text" name="lastname">
      <input type="submit" name="submit" value="Submit">
    </form>
  </body>
</html>
```

## When to use POST?

- POST method is used if the form data contains sensitive or personal information.
- This method doesn't display the submitted form data in the page address field.
- POST has no size limitations, and can be used to send large amount of data.
- Form submissions with POST cannot be bookmarked.

- **The Name Attribute:**

- Each input field must have a name attribute to be submitted.
- If the name attribute is omitted, data of that input field will not be sent to all.
- The example below only submit the "lastname" input field:

- **Example:**

```
<html>
<body>
  <form action="action_page.php" target="_blank" method="post">
    Firstname:<input type="text" value="jhon">
    Lastname:<input type="text" name="lastname" value="harry">
    <input type="submit" name="submit" value="Submit">
  </form>
</body>
</html>
```

- **The Enctype Method:**

- This method specifies encoding of the submitted data(default is url-encoded).

```
<form action="action_page.php" target="_blank" method="post"
      enctype="multipart/form-data">
```

- **<textarea>:** Defines multiline input control.

**Example:**

```
<!DOCTYPE html>
<html> <body>
  <form action="action_page.php">
    <textarea name="message" rows="10" cols="10">Hello good morning
    Have a nice Day</textarea>
    <input type="submit">
  </form>
</body>
</html>
```

The rows attribute specifies the visible number of lines in a text area. The cols attribute specifies the visible width of the text area.

- **<button>**: Defines a clickable button.

**Example:**

```
<html>
  <body>
    <button type="button" onclick="alert('Hello Wolrd!')">Click Me</button>
  </body>
</html>
```

- **<option>**: Defines an option in a drop-down list. By default, the first item in the drop\_down list is selected.

**Example:**

```
<html>
  <body>
    <form action="action_page.php">
      <select name="cars">
        <option value="Volvo">Volvo</option>
        <option value="Fiat">Fiat</option>
        <option value="Audi">Audi</option>
      </select>
      <input type="submit">
    </form>
  </body>
</html>
```

- **<label>**: Defines a label for an <input> element.

**Example:**

```
<html>
  <body>
    <table>
      <tr>
        <td><label>Firstname</label></td>
        <td><input type="text" name="firstname"></td>
      </tr>
    </table>
  </body>
</html>
```

### 3.4 Frames:

- HTML frames are used to divide your browser window into multiple sections where each section can load a separate HTML document. A collection of frames in the browser window is known as a frameset. The window is divided into frames in a similar way the tables are organized: into rows and columns.
- To use frames on a page we use <frameset> tag instead of <body> tag. The <frameset> tag defines, how to divide the window into frames. The **rows** attribute of <frameset> tag defines horizontal frames and **cols** attribute defines vertical frames. Each frame is indicated by <frame> tag and it defines which HTML document shall open into the frame.
- **NOTE : The <frame> tag deprecated in HTML 5.**
- **Example:**

```
<!DOCTYPE html>
<html>
    <head>
        <title>HTML Frames</title>
    </head>
    <frameset rows = "10%,80%,10%">
        <frame name = "top" src = "top_frame.htm" />
        <frame name = "main" src = "main_frame.htm" />
        <frame name = "bottom" src = "bottom_frame.htm" />
    <noframes>
        <body>Your browser does not support frames.</body>
    </noframes>
</frameset>
</html>
```

- **Example :**

```
<!DOCTYPE html>
<html>
    <head>
        <title>HTML Target Frames</title>
    </head>
    <frameset rows = "10%, *,10%" frameborder="1" >
        <frame src = "header.html" name = "header_page" scrolling = "no" >
        <frameset cols = "15%, *" frameborder="1">
```

```
<frame src = "menu.html" name = "menu_page" />
<frame src="main.html" name="main_page" >
</frameset>
    <frame src = "footer.html" name = "footer_page" scrolling = "no" >
<noframes>
    <body>Your browser does not support frames.</body>
</noframes>
</frameset>
</html>
```

### **menu.html**

```
<!DOCTYPE html>
<html>
  <head>
    <title>Menu</title>
  </head>
  <body>
    <body bgcolor = "white">
      <a href = "menu1.html" target = "main_page">Menu1</a>
      <br />
      <br />
      <a href = "menu2.html" target = "main_page">Menu2</a>
      <br />
      <br />
      <a href = "#" target = "main_page">BBC News</a>
    </body>
  </html>
```

### 3.5 Image tags:

- **<img>**: Defines an image.

**Syntax:** `</img>`

- The source file(src) specifies the filename of image source.
- Alternative text(alt) is used when an image cannot be displayed.
- Width and Height of image .

**Example:**

```
<!DOCTYPE html>
```

```
<html>
```

```
  <body>
```

```
    <img src= "image.jpg" alt ="image.com  width=100 height=100>
```

```
  </body>
```

```
</html>
```

### 3.6 Audio/Video:

- HTML5 features, include native audio and video support without the need for Flash.
- The HTML5 `<audio>` and `<video>` tags make it simple to add media to a website. You need to set **src** attribute to identify the media source and include a **controls** attribute so the user can play and pause the media.

- **Video tag :**

- Here is the simplest form of embedding a video file in your webpage:-

```
<video src="foo.mp4" width="300" height="200" controls>
```

Your browser does not support the `<video>` element.

```
</video>
```

- The current HTML5 draft specification does not specify which video formats browsers should support in the video tag. But most commonly used video formats are –

- **Ogg** – Ogg files with Theora video codec and Vorbis audio codec.
- **mpeg4** – MPEG4 files with H.264 video codec and AAC audio codec.

- You can use `<source>` tag to specify media along with media type and many other attributes. A video element allows multiple source elements and browser will use the first recognized format.

➤ **Example:**

```

<!DOCTYPE HTML>

<html>

  <body>

    <video width="300" height="200" controls autoplay>
      <source src="/html5/foo.ogg" type="video/ogg" />
      <source src="/html5/foo.mp4" type="video/mp4" />
    </video>
  </body>
</html>

```

- The HTML5 video tag can have a number of attributes to control the look and feel and various functionalities of the control

| Attribute  | Description                                                                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| autoplay   | This boolean attribute if specified, the video will automatically begin to play back as soon as it can do so without stopping to finish loading the data. |
| autobuffer | This boolean attribute if specified, the video will automatically begin buffering even if it's not set to automatically play.                             |
| controls   | If this attribute is present, it will allow the user to control video playback, including volume, seeking, and pause/resume playback.                     |
| height     | This attribute specifies the height of the video's display area, in CSS pixels.                                                                           |
| Loop       | This boolean attribute if specified, will allow video automatically seek back to the start after reaching at the end.                                     |
| preload    | This attribute specifies that the video will be loaded at page load, and ready to run. Ignored if autoplay is present.                                    |
| poster     | This is a URL of an image to show until the user plays or seeks.                                                                                          |
| Src        | The URL of the video to embed. This is optional; you may instead use the <source> element within the video block to specify the video to embed            |
| width      | This attribute specifies the width of the video's display area, in CSS pixels.                                                                            |

- **Audio tag :**

- HTML5 supports <audio> tag which is used to embed sound content in an HTML or XHTML document as follows.

```
<audio src="foo.wav" controls autoplay>
```

Your browser does not support the <audio> element.

```
</audio>
```

- The current HTML5 draft specification does not specify which audio formats browsers should support in the audio tag. But most commonly used audio formats are **ogg**, **mp3** and **wav**.
- You can use <source> tag to specify media along with media type and many other attributes. An audio element allows multiple source elements and browser will use the first recognized format.

- **Example:**

```
<!DOCTYPE HTML>
```

```
<html> <body>
```

```
  <audio controls autoplay>
```

```
    <source src="/html5/audio.ogg" type="audio/ogg" />
```

```
    <source src="/html5/audio.wav" type="audio/wav" />
```

Your browser does not support the audio element.

```
  </audio>
```

```
</body>
```

```
</html>
```

- The HTML5 audio tag can have a number of attributes to control the look and feel and various functionalities of the control:

| Attribute  | Description                                                                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Autoplay   | This boolean attribute if specified, the audio will automatically begin to play back as soon as it can do so without stopping to finish loading the data. |
| autobuffer | This boolean attribute if specified, the audio will automatically begin buffering even if it's not set to automatically play.                             |
| Controls   | If this attribute is present, it will allow the user to control audio playback, including volume, seeking, and pause/resume playback.                     |
| Loop       | This boolean attribute if specified, will allow audio automatically seek back to the start after reaching at the end.                                     |
| Preload    | This attribute specifies that the audio will be loaded at page load, and ready to run. Ignored if autoplay is present.                                    |
| Src        | The URL of the audio to embed. This is optional; you may instead use the <source> element within the video block to specify the video to embed            |

### 3.7 Link Tags:

- In HTML link tags are defined with <a> tag.
- <a>: Defines a hyperlink. When we click on a link it will jump to another document when we move the mouse over a link ,a mouse will turn into a little hand.

**Syntax:** < a href="document location"> </a>

The links address or destination is specified in the **href** attribute.

**Example:**

```
<!DOCTYPE html>
<html>
  <body>
    <a href="a.html">This is link</a>
  </body>
</html>
```

### 3.8 List Tags:

- HTML offers three ways for specifying list of information .All list must contain Or more list elements . List may contain:

➤ **<ul>:** Defines an Unordered List. This will list items using plain bullets.

Unordered list is a collection of related items that have no special or sequence.

- **Example:**

```
<!DOCTYPE html>
<html> <head>
    <title>HTML Unordered List</title>
</head>
<body>
    <ul>
        <li>Mango</li>
        <li>Grapes</li>
        <li>Apple</li>
    </ul>
</body>
</html>
```

- We can use **type** attribute for <ul> tag to specify the type of bullet we like. By default it is disc. Following are the possible options:

```
<ul type="square">
<ul type="disc">
<ul type="circle">
```

- **Example:**

```
<!DOCTYPE html>
<html> <head>
    <title>HTML Unordered List</title>
</head>
<body> <ul type="square">
    <li>Mango</li>
    <li>Grapes</li>
    <li>Apple</li>
</ul>
</body>
</html>
```

- **<ol>:** Defines an Ordered List. This will use different schemes of numbers to list items. We use ordered list if items to be displayed in numbers instead of bulleted list. The numbering starts at one and is incremented by one for each successive ordered list element tagged with <li>.

▪ **Example:**

```
<!DOCTYPE html>
<html>
  <head> <title>HTML Ordered List</title>
  </head>
<body> <ol> <li>Mango</li>
      <li>Grapes</li>
      <li>Apple</li>
    </ol>
</body>
</html>
```

- We can use **type** attribute for <ol> tag to specify type of numbering we like. By default it is number. Following are the possible options.

- ❖ <ol type="1"> - Default case Numerals.
- ❖ <ol type="I"> - Upper-Case Roman Numerals.
- ❖ <ol type="i"> - Lower-Case Roman Numerals.
- ❖ <ol type="a"> - Lower Case Letters.
- ❖ <ol type="A"> -Upper Case Letters.

▪ **Example:**

```
<!DOCTYPE html>
<html>
  <head> <title>HTML Ordered List</title> </head>
  <body>
    <ol type="I">
      <li>Mango</li>
      <li>Grapes</li>
      <li>Apple</li>
    </ol>
  </body>
</html>
```

- **<dl>:** Defines a Definition List. HTML and XHTML supports list style which is called Definition lists where entries are listed like in a dictionary or encyclopedia. A definition list is a way to present a glossary, list of terms, or other name/value list.

- Definition List makes use of following three tags:
- `<dl>` - Defines the start of the list.
- `<dt>` - A term.
- `<dd>` - Term Definition.
- `</dl>` - Defines the End of the list.

▪ **Example:**

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Definition List</title>
  </head>
  <body>
    <dl>
      <dt>HTML</dt>
      <dd>This stands for Hyper Text Markup Language.</dd>
      <dt>HTTP</dt>
      <dd>This stands for Hyper Text Transfer Protocol.</dd>
    </dl>
  </body>
</html>
```

### 3.9 Table Tags:

- The HTML <table> tag is used for defining a table. The table tag contains other tags that define the structure of the table.
- <table>: Defines a table
- <Caption>: Defines table caption (specifies name to a table)
- <th>: Defines a header cell in a table
- <tr>: Defines a row in a table
- <td>: Defines a cell in a table
- <thead> - Table head
- <tbody> - Table body

**Example:**

```
<!DOCTYPE html>
<html>
    <head>
        <title>Table</title>
    </head>
    <body>
        <table>
            <thead>
                <tr> <th>Month</th> <th>Savings</th> </tr>
            </thead>
            <tbody>
                <tr> <td>January</td> <td>$100</td> </tr>
                <tr> <td>February</td> <td>$80</td> </tr>
            </tbody>
        </table>
    </body>
</html>
```

- Following code:

```
<table>
    <tr> <td>Row1-Col1</td> <td>Row1-Col2</td> </tr>
    <tr> <td>Row2-Col1</td> <td>Row2-Col2</td> </tr>
</table>
```

Produces a table that looks like:

Row 1 - Col 1   Row 1 - Col 2

Row 2 - Col 1   Row 2 - Col 2

- Inside the **<table>** tag are **<tr>** or table row tags which in turn contain **<td>** or table cell tags.
- Data goes in the **<td>** cell. That data can be text, other tags or whatever. Remember that first comes **<table>**, then **<tr>**, then **<td>**, then *and only then*, data.
- Below I added some **CSS** and the border attribute (border="1") to better display the rows and columns.

|               |               |
|---------------|---------------|
| Row 1 - Col 1 | Row 1 - Col 2 |
| Row 2 - Col 1 | Row 2 - Col 2 |

- The HTML **<table>** tag also supports the following additional attributes –

| Attribute   | Value                                      | Description                                                                          |
|-------------|--------------------------------------------|--------------------------------------------------------------------------------------|
| Abbr        | abbreviated_text                           | <i>Deprecated</i> – Specifies an abbreviated version of the content in a cell.       |
| Align       | right<br>left<br>center<br>justify<br>char | <i>Deprecated</i> – Visual alignment.                                                |
| bgcolor     | rgb(x,x,x)<br>#hexcode<br>colorname        | <i>Deprecated</i> – Specifies the background color of the table.                     |
| border      | Pixels                                     | <i>Deprecated</i> – Specifies the border width. A value of "0" means no border.      |
| cellpadding | pixels or %                                | <i>Deprecated</i> – Specifies the space between the cell borders and their contents. |
| cellspacing | pixels or %                                | <i>Deprecated</i> – Specifies the space between cells.                               |

|         |                                                                           |                                                                                                                                                                   |
|---------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frame   | void<br>above<br>below<br>hsides<br>lhs<br>rhs<br>vsides<br>box<br>border | <i>Deprecated</i> – Used in conjunction with the border attribute, specifies which side of the frame that makes up the border surrounding the table is displayed. |
| Rules   | none<br>groups<br>rows<br>cols<br>all                                     | <i>Deprecated</i> – Used in conjunction with the border attribute, specifies which rules appear between the cells of the table.                                   |
| summary | Text                                                                      | <i>Deprecated</i> – Specifies the summary of the content.                                                                                                         |
| Width   | pixels or %                                                               | <i>Deprecated</i> – Specifies the width of the table.                                                                                                             |

- Here's the rub. There has to be the same number of cells in each row *unless* you indicate a table cell spans multiple columns and/or rows.
- Multiple columns:

```
<table border="1">
  <tr>
    <td colspan="2">Row1-Col1 & Col2</td>
  </tr>
  <tr>
    <td>Row 2 - Col 1</td>
    <td>Row 2 - Col 2</td>
  </tr>
</table>
```

|                       |               |
|-----------------------|---------------|
| Row 1 - Col 1 & Col 2 |               |
| Row 2 - Col 1         | Row 2 - Col 2 |

- Multiple rows:

```
<table border="1">
<tr>
  <td>Row 1 - Col 1</td>
  <td rowspan="2">Row 1 & Row 2 - Col 2</td>
</tr>
<tr>
  <td>Row 2 - Col 1</td>
</tr>
</table>
```

|               |                       |
|---------------|-----------------------|
| Row 1 - Col 1 | Row 1 & Row 2 - Col 2 |
| Row 2 - Col 1 |                       |

- or a combination of both:

```
<table border="1">
<tr>
  <td>Row 1 - Col 1</td>
  <td>Row 1 - Col 2</td>
  <td>Row 1 - Col 3</td>
  <td rowspan="4">Row 1, 2, 3 & 4 - Col 4</td>
</tr>
<tr>
  <td>Row 2 - Col 1</td>
  <td colspan="2" rowspan="3">Row 2, 3 & 4 - Col 2 & 3</td>
</tr>
<tr>
  <td>Row 3 - Col 1</td>
</tr>
<tr>
  <td>Row 4 - Col 1</td>
</tr>
```

```
<tr>
  <td colspan="2">Row 5 - Col 1 & 2</td>
  <td colspan="2">Row 5 - Col 3 & 4</td>
</tr>
<tr>
  <td colspan="4">Row 6 - Col 1, 2, 3 & 4</td>
</tr>
</table>
```

|                         |                          |                   |                         |
|-------------------------|--------------------------|-------------------|-------------------------|
| Row 1 - Col 1           | Row 1 - Col 2            | Row 1 - Col 3     | Row 1, 2, 3 & 4 - Col 4 |
| Row 2 - Col 1           | Row 2, 3 & 4 - Col 2 & 3 |                   |                         |
| Row 3 - Col 1           |                          |                   |                         |
| Row 4 - Col 1           |                          |                   |                         |
| Row 5 - Col 1 & 2       |                          | Row 5 - Col 3 & 4 |                         |
| Row 6 - Col 1, 2, 3 & 4 |                          |                   |                         |

### 3.10 Styles and Semantic Tags:

- **<Style>**: Defines an style information for a document

**Syntax:** <tag name style="property:value;"></tag name>

Property is css property and Value is css value.

**Example:**

```
<!DOCTYPE html>
<html>
  <body>
    <h1 style="text-align:"center;">Hello</h1>
  </body>
</html>
```

- **<div>**: The HTML <div> tag is used for defining a section of your document. With the div tag, you can group large sections of HTML elements together and format them with CSS.

**Syntax:** <div>Contents goes here.....</div>

**Example:**

```
<!DOCTYPE html>
<html>
  <body>
    <p>This is some text.</p>
    <div style="color:#0000FF">
      <h3>This is a heading in a div element</h3>
      <p>This is some text in a div element.</p>
    </div>
    <p>This is some text.</p>
  </body>
</html>
```

### 3.11 Programming Tags:

- **<script>**: Defines an client-side script. <script> element either contains scripting Statements, or it points to an external script file through the src attribute Javascript can be used for image manipulation , form validation, and Dynamic changes of content .To select an HTML element, Javascript often uses document.getElementById(id) method.

**Syntax:**<script>Contents Goes</script>

**Example:**

```
<!DOCTYPE html>
<html>
<body>
<p id="demo"></p>
<script>
    document.getElementById("demo").innerHTML = "Hello JavaScript!";
</script>
</body>
</html>
```